MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular Examination November 2025
First Semester
Skill Development Components (SDC) – MG1SDCSNP101 – HUMAN ANATOMY
(2025 Admission Onwards)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

ANSWER SCHEME

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
One Word Questions
Answer all questions
Each question carries 1 mark

1. **Which tissue connects muscles to bones?** **(CO1 / K)**
   Ans: Tendon
2. **Which plane divides the body into upper and lower halves?** **(CO1 / K)**
   Ans: Transverse (horizontal) plane
3. **What is the outer covering of bone called?** **(CO1 / K)**
   Ans: Periosteum
4. **Which muscle type is voluntary?** **(CO1 / U)**
   Ans: Skeletal muscle
5. **What is the function of the endocardium?** **(CO2 / U)**
   Ans: Lines the heart chambers and reduces friction
6. **What is respiratory unit?** **(CO2 / K)**
   Ans: Alveolus along with respiratory bronchioles and ducts
7. **Name the main bronchi.** **(CO2 / K)**
   Ans: Right and Left primary bronchi
8. **Name the structures that form the brainstem.** **(CO3 / K)**
   Ans: Midbrain, Pons, Medulla oblongata
9. **What is the function of the thyroid gland?** **(CO3 / U)**
   Ans: Regulates metabolism by secreting T3 and T4
10. **Name the structure connecting the pharynx to the stomach.** **(CO4 / U)**
    Ans: Oesophagus

(1×10 = 10 marks)

Part B
Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

11. **Classify joints with suitable examples.** **(CO1 / K)**

- Fibrous joint – Sutures
- Cartilaginous joint – Intervertebral disc
- Synovial joint – Knee, shoulder

12. **Describe the articulating surfaces of the elbow joint.** **(CO1 / U)**

- Humerus: trochlea & capitulum
- Ulna: trochlear notch
- Radius: head of radius

13. **Describe the structure of a ball-and-socket joint.** **(CO1 / U)**

- Rounded head fits into cup-like socket
- Multiaxial movements (flexion, extension, rotation)
- Example: Hip/shoulder

14. **Discuss the ligaments in knee joint.** **(CO1 / U)**

- ACL, PCL
- MCL, LCL
- Patellar ligament
- Meniscal ligaments

15. **Describe the paranasal air sinuses.** **(CO2 / U)**

- Frontal, Maxillary, Ethmoidal, Sphenoidal
- Functions: lighten skull, resonance, warm/moisten air

16. **Apply knowledge of respiratory muscles to explain breathing movements. (CO2 / A)**

- **Inspiration:** diaphragm contracts; external intercostals raise ribs
- **Forced expiration:** internal intercostals & abdominal muscles

17. **Describe the gross anatomy of lungs and their lobes.** **(CO2 / U)**

- Right lung: 3 lobes
- Left lung: 2 lobes + cardiac notch
- Apex, base, hilum

18. **Identify anatomical location of pancreas in relation to duodenum & stomach. (CO3 / A)**

- Retroperitoneal
- Head in C-loop of duodenum
- Body behind stomach
- Tail near spleen

19. **Explain the functions of the autonomic nervous system. (CO3 / U)**

- Controls involuntary activities
- Sympathetic → fight/flight
- Parasympathetic → rest/digest

20. **Describe the structure and functions of the spinal cord. (CO3 / U)**

- Central grey matter; peripheral white matter
- Conducts nerve impulses
- Reflex center

21. **Compare right and left lobes of liver. (CO4 / U)**

- Right lobe: larger, broader
- Left lobe: smaller, narrow

22. **Describe the gross anatomy of kidneys. (CO4 / U)**

- Bean-shaped retroperitoneal organs
- Outer cortex, inner medulla
- Renal pyramids, pelvis, hilum

(2×10 = 20 marks)

Part C
Short Essay Questions
Answer any 5 questions
Each question carries 4 marks

23. **Describe the types of movements possible at synovial joints with suitable examples. (CO1 / U)**

- Flexion/Extension – elbow
- Abduction/Adduction – hip
- Rotation – atlas-axis
- Circumduction – shoulder
- Gliding – intercarpal joints

24. **Describe the types of bones based on shape with examples. (CO1 / U)**

- Long – femur
- Short – carpals
- Flat – scapula
- Irregular – vertebra
- Sesamoid – patella

25. **Describe the gross and microscopic structure of lungs. (CO2 / U)**
Gross: lobes, fissures, hilum, pleura
Microscopic: bronchi → bronchioles → alveoli; type I & type II pneumocytes; capillaries

26. **Explain the flow of blood through the heart. (CO2 / U)**

- SVC/IVC → RA → tricuspid → RV
- Pulmonary artery → lungs
- Pulmonary veins → LA → mitral → LV → aorta

27. **Explain the cranial nerves and their functions. (CO3 / U)**

- I smell, II vision
- III/IV/VI eye movements
- V mastication, facial sensation
- VII facial expression
- VIII hearing & balance
- IX/X swallowing & autonomic
- XI shoulder movement
- XII tongue movements

28. **Describe the parts of the stomach and their functions. (CO4 / U)**

- Cardia – entry
- Fundus – gas storage
- Body – mixing
- Pylorus – regulates emptying
- Secretes HCl, mucus, enzymes

29. **Evaluate the structural importance of pancreatic islets. (CO4 / E)**

- Islets contain α, β, δ, PP cells
- Endocrine: insulin, glucagon, somatostatin
- Rich vascularity → rapid hormone release
- Maintains glucose homeostasis

(4×5 = 20 marks)